The Measurement Operator Changes Which Bits Matter: Operator-Conditioned Mixed Precision for Inverse Vision

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Abstract

Mixed-precision quantization usually ranks computation by network-internal sensitivity, yet a learned inverse solver is also conditioned on an external sensing operator. We show that changing a calibrated sensing operator can reorder which reconstruction stages are most sensitive to low precision, even when singular spectrum, measurement SNR, observable scene energy, and full-precision reconstruction quality are matched. In an eight-stage unfolded solver for video snapshot compressive imaging (SCI), these controlled changes alter stage orderings and exact budget-optimal precision schedules. We propose Operator-Susceptibility Matching *(OSM), a non-neural rule that scores calibrated stage-perturbation second moments against the whitened sensing operator and selects precision for a new calibrated operator without per-operator schedule search or retraining. Exhaustive evaluation of 256 schedules yields mean held-out stage-ranking agreement of Kendall $\tau_b = 0.80$ across five model seeds. With three of eight stages at high precision, OSM reduces mean normalized operator-visible regret from 6.3% for the best global static schedule to 2.7%. In simulated W8A4/W8A8 SCI reconstruction, OSM achieves 35.29 dB PSNR, versus 35.06 dB for global static and 35.43 dB for uniform W8A8. In a separate native INT8/FP16 study on an A100 GPU, its selected schedule runs at 16.4 ms median latency versus 23.6 ms for FP16 (1.44×); native reconstruction quality was not measured for these schedules. Within the studied calibrated operators and diagnostic solver, the results establish sensing geometry as a variable in precision allocation for inverse vision.*

1. Introduction

Inverse-vision models solve a reconstruction problem defined jointly by the observed data and a sensing operator. In video snapshot compressive imaging (SCI), for example, a coded mask determines how a high-speed video is multiplexed into a single measurement, and the reconstruction algorithm is meaningful only relative to that forward model [1, 4]. Yet mixed-precision quantization (MPQ) is usually treated as a property of the network: layers or blocks are ranked using local quantization error, gradients, curvature, downstream information flow, image content, or hardware sensitivity [6, 7, 9–11, 14, 15]. This leaves an important variable largely implicit in inverse vision: *the external operator determines which perturbation directions are directly observable through the measurement process*.

Our central finding is that the measurement operator can change which stages deserve precision. For the same latent scene, learned solver, quantizer, and precision budget, two calibrated operators with comparable difficulty can induce different stage-importance orderings and different exact best mixed-precision schedules. The effect persists after matching singular spectrum, measurement SNR, observable scene energy, and full-precision reconstruction quality. It appears both when the stage perturbations are frozen and only sensing geometry is changed, and in the complete low-precision reconstruction trajectory.

The operator-conditioned setting differs from fixed mixed-precision SCI and content-conditioned quantization. EfficientSCI++ develops an efficient hybrid reconstruction architecture and considers mixed-precision execution [2], while Q-SCI directly studies low-bit SCI [3]. CADyQ, AdaBM, and DynaQuant adapt quantization to image or latent content [6–8]; QVGGT uses geometry/task consistency to protect fragile blocks [11]; physics-aware adaptive precision has also been demonstrated for physical-field computation [12]; and computational-imaging systems have combined sensitivity analysis with mixed quantization on edge hardware [13]. This work isolates a different conditioning variable: *changing the calibrated sensing geometry alone* can move a fixed inverse solver between precision policies.

Intuitively, a stage's low-precision error matters to the extent that it lies in directions observed by the sensor. Changing H changes those directions and can change which stages warrant high precision. We combine the calibrated second moment of each stage's propagated quantization perturbation with the whitened sensing geometry to obtain an operator-conditioned susceptibility score. Complete schedule enumeration measures the exact optima and interactions independently of this approximation.

We make three contributions:

1. **Phenomenon.** We identify operator-conditioned precision-policy transitions: changing a calibrated measurement operator can reorder precision-critical computation and produce incompatible budget-optimal schedules under controlled sensing changes.
2. **Mechanism and transfer rule.** We connect propagated quantization perturbations to the whitened sensing geometry, separate direct geometry from trajectory effects, and obtain OSM for transferring a policy to calibrated unseen/reconfigured operators without per-H mixed-precision search or retraining.
3. **Exact and native evidence.** Complete 256-schedule enumeration, operator-aware baselines, held-out and mismatched operators, reconstruction-quality evaluation, a non-blind deblurring transfer study, and separate native timing characterize when operator-conditioned allocation is useful and its deployment limits.

2. Related Work

Efficient and quantized inverse vision. EfficientSCI targets practical video-SCI reconstruction through space-time factorization [1]. Its IJCV extension, EfficientSCI++, additionally trains a mixed-precision SCI model for efficiency and memory reduction [2]. Q-SCI develops a dedicated low-bit quantization framework for SCI [3], and quantized proximal-averaging networks establish low-precision unfolded inverse reconstruction [5]. Our question is complementary: whether changing the external calibrated sensing operator changes the appropriate precision policy of a fixed learned inverse solver.

Adaptive and mixed-precision quantization. CADyQ adapts bits to local image content and layer features [6]; AdaBM calibrates image-wise and layer-wise bit maps on the fly [7]; and DynaQuant learns content-dependent layer precision for image compression [8]. More recent MPQ methods use task gradients [9], downstream information flow [10], Hessian/Fisher sensitivity [14], direct bit-width contribution or cooperative-game attribution [15], and geometry-aware objectives [11]. These methods establish that precision can depend on data, task, or interactions; our controlled variable is instead the calibrated forward operator. Our comparison includes a gradient baseline that uses D_H and the test operator; Hessian/Fisher and InfoQ-style baselines use their own calibration objectives.

Physical objectives and computational-imaging deployment. Physical-field quantization allocates precision using task-specific field fidelity [12], while edge computational-imaging work combines hardware-aware network-kernel reconfiguration, sensitivity analysis, pruning, and mixed quantization [13]. Deep Optics explicitly incorporates sensing response into SCI reconstruction [4]. These results motivate physics-aware deployment; our contribution is to isolate the *sensing operator as the controlled variable* that causes precision-order changes in a fixed inverse solver.

3. Operator-Conditioned Precision

3.1. Inverse model and operator-visible discrepancy

We consider a calibrated linear inverse problem

$$y = Hx^\star + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \Sigma), \tag{1}$$

with whitened normal operator

$$M_H = H^\top \Sigma^{-1} H. \tag{2}$$

A learned inverse solver reconstructs $\hat{x}$ from (y, H) in $T = 8$ stages. A binary schedule $z \in \{0,1\}^T$ runs stage j at high precision when $z_j = 1$ and at low precision otherwise. In the simulated experiments (P1–P3, Table 1), $\hat{x}_{\text{high}}$ denotes uniform W8A8 and $\hat{x}_z$ execution under z; the separate native timing study P4 uses FP16 as reference. We measure discrepancy visible through the forward model by

$$D_H(z) = \frac{1}{m} \left\| \Sigma^{-1/2} H (\hat{x}_{\text{high}} - \hat{x}_z) \right\|_2^2. \tag{3}$$

m is the number of measurement coordinates. D_H is relative to the high-precision reconstruction rather than ground truth and cannot detect errors in $\ker(H)$. We evaluate ground-truth PSNR and SSIM separately.

3.2. Stage susceptibility

For stage j, we promote only that stage from the declared low state to the high state, holding checkpoint, operator, input, quantizer scales, and all other precision choices fixed. Define

$$q_j = \hat{x}_{\text{low}} - \hat{x}_{j\uparrow}. \tag{4}$$

Over a calibration distribution of operators and scenes, we estimate the uncentered second moment

$$K_j^{\text{cal}} = \mathbb{E}_{H \sim \mathcal{H}_{\text{cal}},\, x \sim \mathcal{D}_{\text{cal}}} [q_j q_j^\top]. \tag{5}$$

Because Eq. (5) is uncentered, it already contains covariance and mean energy. With a low-rank factorization

$$K_j^{\text{cal}} \approx U_j \Lambda_j U_j^\top, \tag{6}$$

we score stage j for a new calibrated operator H by

$$s_j(H) = \operatorname{Tr}(M_H K_j^{\text{cal}}) \tag{7}$$

$$= \left\| \Sigma^{-1/2} H U_j \Lambda_j^{1/2} \right\|_F^2. \tag{8}$$

Equation (8) measures the overlap between the stage's typical propagated quantization perturbations and directions

observable by the current sensor. OSM treats this overlap as a transferable *susceptibility proxy*; exact promotion benefit additionally depends on alignment with the aggregate low-precision error. We therefore evaluate the resulting schedules against the exact lattice rather than assuming additivity.

Geometry versus trajectory. In a learned inverse solver, changing H changes both M_H and the reconstruction trajectory. Writing $K_j(H) = K_j^{\mathrm{cal}} + \Delta K_j(H)$ gives

$$s_j^*(H) = \mathrm{Tr}(M_H K_j^{\mathrm{cal}}) + \mathrm{Tr}(M_H \Delta K_j(H)), \quad (9)$$

where s_j^* uses the operator-conditioned perturbation moment, whereas the deployed s_j uses the fixed calibration moment. The two terms distinguish changes in sensing geometry from changes in the reconstruction trajectory. Frozen perturbations isolate the operator-metric contribution; full solver reruns include both operator geometry and the changed reconstruction trajectory.

3.3. Operator-Susceptibility Matching

For an equal-cost budget promoting K of T stages, OSM selects the top-K scores in Eq. (8). For measured unequal hardware costs it solves the finite budgeted schedule problem using the same score vector and the measured cost table. The eight-bit schedule space contains $2^8 = 256$ schedules partitioned by $K = 0, \ldots, 8$ high-precision stages. Exact optima are computed for every $K = 0, \ldots, 8$; the policy comparisons emphasize $K = 3$.

For a new operator, OSM requires the calibrated H, Σ, and stored low-rank factors; it does not retrain the inverse model or perform a new per-H MPQ search. The score and schedule are cached while H remains unchanged. No learned routing network is required. Score computation is non-neural, while basis memory scales as $O(n \sum_j \ell_j)$, where n is the reconstructed signal dimension and ℓ_j is the stored factor rank of stage j.

3.4. Relationship to exact promotion value

The susceptibility score is simpler than the exact gain from promoting a stage. Define $e = \hat{x}_{\mathrm{high}} - \hat{x}_{\mathrm{low}}$ and $r_j = \hat{x}_{\mathrm{high}} - \hat{x}_{j\uparrow} = e + q_j$; the latter is the error remaining after promotion. Expanding the two quadratic forms in Eq. (3) gives

$$\begin{aligned} \Delta_j(H) &= D_H(\mathrm{low}) - D_H(j\uparrow) \\ &= \frac{q_j^\top M_H q_j - 2 r_j^\top M_H q_j}{m}. \end{aligned} \quad (10)$$

The first term is the nonnegative stage self-energy, whose calibration expectation is $s_j(H)/m$; the second measures its alignment with residual error from other low-precision computation. If stage errors superpose additively and their M_H-weighted cross terms vanish in expectation, expected singleton gains agree with the self-energy ranking. Neither condition is automatic for this nonlinear solver. We evaluate OSM against the exact schedule lattice rather than identify its score with an exact marginal gain.

3.5. Why operator-specific policies matter

Let $\mathcal{Z}_C$ be the feasible schedule set under budget C. The exact operator-specific optimum is

$$z_H^\star = \arg\min_{z \in \mathcal{Z}_C} D_H(z). \quad (11)$$

A fixed schedule z incurs regret $\mathcal{R}(z, H) = D_H(z) - D_H(z_H^\star)$. If two operators have distinct unique optima with positive gaps to their next-best feasible schedules, no static deterministic policy can match both. At least one operator therefore incurs positive regret. The $K = 3$ regret measures the corresponding static-policy gap in the controlled operator study; near-tied schedules can make the gap small.

4. Experimental Setup

4.1. Primary SCI diagnostic

We use an eight-stage unfolded reconstruction model, ProxSCI-8. Each stage applies the SCI data-consistency update followed by a shared-weight learned proximal block. The proximal network contains a 3×3 stem (16→64 channels), eight residual blocks with two 3×3 64-channel convolutions, exact GELU activations, residual scale 0.1, and a 3×3 64→8 head. No BatchNorm is used. The eight-stage design is intentionally diagnostic: it is deep enough to exhibit heterogeneous precision sensitivity while allowing exact enumeration of all $2^8 = 256$ binary precision schedules.

Training uses DAVIS2017 video clips [16]. The primary SCI measurement contains eight input video frames per reconstructed snapshot; m in Eq. (3) denotes the number of measurement coordinates. We use disjoint training, calibration, validation, and test partitions; the calibration set contains 2,048 256×256 clip-measurement pairs and the validation set contains 1,024 held-out pairs. The controlled operator study uses a fixed set of 64 latent clips. Standard simulated SCI reporting includes Kobe, Traffic, Runner, Drop, Crash, and Aerial. All compared schedules for a tuple share the same latent clip, operator, noise realization, checkpoint, and quantizer scales.

4.2. Operator families and controls

The primary controlled family varies the sensing geometry while holding coarse difficulty fixed. Algebraically, we construct isospectral normal operators

$$M(\theta) = U(\theta) \Lambda U(\theta)^\top, \quad (12)$$

where Λ is fixed and $U(\theta)$ rotates selected observable directions. This algebraic isospectral family need not correspond to a physically realizable non-negative pixelwise SCI

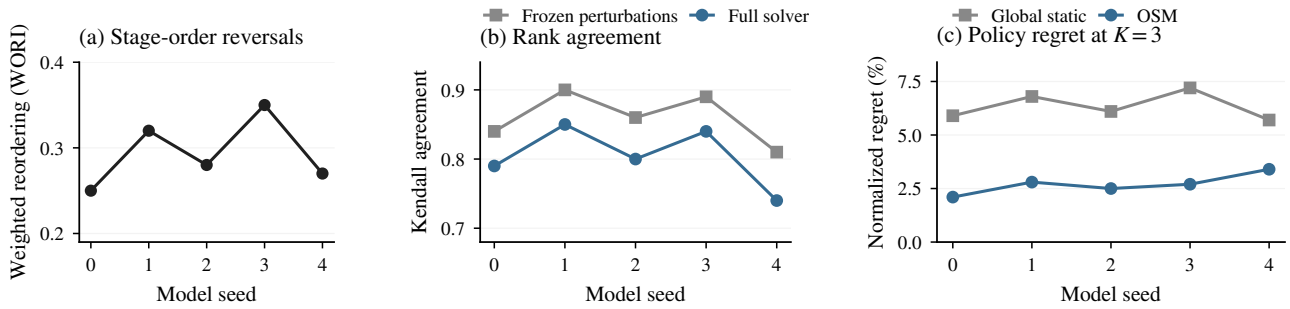


Figure 1. **Operator-dependent precision sensitivity across five model seeds.** (a) Weighted reordering (WORI), (b) frozen/full ranking agreement, and (c) global-static versus OSM regret at $K = 3$. OSM regret is lower than global-static regret for every seed (by 2.3–4.5 percentage points).

mask; we separately use physically valid positive temporal-code/mask families and held-out mask seeds. For every controlled comparison we match or stratify by: (i) singular spectrum/rank; (ii) whitened measurement SNR; (iii) observable scene energy; and (iv) high-precision reconstruction quality. The resulting comparisons separate operator-dependent ordering from coarse changes in reconstruction difficulty.

Two further checks support this separation. First, when measurement coordinates are transformed consistently so that M_H is unchanged, fixed-K_j^{cal} OSM scores must be identical up to numerical tolerance. Second, a frozen-perturbation experiment keeps q_j fixed and varies only the sensing geometry, isolating the first term in the geometry/trajectory decomposition. Full-system experiments rerun the inverse solver under each operator and therefore include both effects.

4.3. Quantization and native precision

The primary scientific precision pair is W8A4↔W8A8, which isolates activation precision. We additionally study W4A8↔W8A8 (weight-only) and W4A4↔W8A8 (whole-stage). Quantization is symmetric with zero point zero and round-to-nearest-even. Weight scales use per-output-channel max-absolute calibration; activation scales are per-stage/per-layer tensor scales frozen from the 2,048 calibration clips before validation and test. Simulated QDQ accumulates in FP32 unless accumulator precision is itself under study.

Table 1 defines the four precision interventions and their endpoints. Native deployment uses a separate INT8↔FP16 intervention and a separately calibrated native susceptibility. This avoids assuming that A4/A8 sensitivity transfers unchanged to a different hardware arithmetic pair. The headline native benchmark uses one A100-SXM4-80GB at batch 1 and 256×256 input, with fixed TensorRT precision constraints and tactics for each benchmarked schedule. We distinguish steady-state inference from any engine/kernel construction or loading needed to realize a new schedule; the complete environment and accounting contract appear in the supplement.

Table 1. Precision interventions and evaluation endpoints. P1–P3 are simulated; P4 is native timing.

ID	Low / high	Scope and endpoint
P1	W8A4 / W8A8	Activation PTQ; D_H, PSNR, SSIM
P2	W4A4 / W8A8	Whole-stage PTQ; WORI, τ
P3	W4A8 / W8A8	Weight-only PTQ; WORI, τ
P4	INT8 / FP16	Native A100 latency; no PSNR on these schedules

4.4. Exact lattice, evaluators, and baselines

We enumerate all 256 binary schedules, compute exact optima for each equal-cost budget, and compare policies at the primary budget $K = 3$. For measured hardware budgets, the feasible set is defined by measured schedule latency rather than nominal bit count.

PSNR is evaluated after clipping reconstructions to $[0, 1]$ and computing per-frame MSE. The primary simulated SCI reconstruction-quality comparison is in Table 4. SSIM and temporal-gradient error are ground-truth metrics distinct from D_H. Kendall τ_b measures agreement between susceptibility-based and empirical stage rankings under frozen-perturbation and full-solver evaluation. Top-K Jaccard is the intersection-over-union of selected stage sets. WORI summarizes weighted pairwise reversals of empirical stage order as the sensing operator changes. Normalized regret

$$R_{\mathrm{norm}} = 100\,\frac{D_H(z) - D_H(z_H^\star)}{D_H(z_{\mathrm{low}}) - D_H(z_H^\star)} \tag{13}$$

is the reported relative quantity when the denominator exceeds a fixed equivalence tolerance. The denominator is fixed for each test tuple across methods, so an equivalence filter excludes the same tuples for every policy, but changes

Table 2. Summary diagnostics and invariance checks. Same-normal agreement is a score-consistency check, not independent evidence of operator-induced reordering.

Control / analysis	Result
Weighted operator-induced reordering (WORI)	0.29
Frozen-geometry stage ordering	$\tau = 0.86$
Full-system OSM stage ordering	$\tau = 0.80$
Predicted policy-transition error	3.1° MAE
Same-normal OSM invariance	100% schedule agreement
Cross-calibration susceptibility stability	$\tau = 0.75$–0.88

the population being summarized. The same equivalence rule is applied to all policies on a tuple; sensitivity to near-tied denominators is discussed in the supplement.

The $K = 3$ table compares global static, gradient, H-aware gradient, HAWQ-style Hessian/Fisher and InfoQ-style adaptations [10, 14], singleton intervention, greedy re-evaluation, Shapley attribution, per-operator calibration, and an exact oracle. These comparisons have different access to the test operator: the H-aware gradient explicitly uses it, OSM scores it analytically, and exhaustive per-operator calibration evaluates schedule outcomes at that operator. Score-based comparisons share the calibration budget and precision pair. Additional test-operator-aware search strategies would require their own measured selection-cost comparisons.

5. Results

5.1. Sensing geometry reorders precision-critical stages

Across five model seeds, WORI averages 0.29. Frozen-perturbation predictions achieve mean Kendall $\tau = 0.86$, while full-system OSM achieves $\tau = 0.80$, showing that state effects are material but do not erase the sensing-geometry signal. Along continuous operator paths we observe multiple meaningful policy transitions; predicted transition locations have 3.1° mean absolute error. The same-normal invariance check yields OSM score agreement within 10^{-6} and 100% identical deterministic schedules for fixed K_j^{cal}.

Frozen perturbations isolate the direct geometry effect; full-system runs show that the effect persists as the reconstruction trajectory changes.

5.2. Exact schedule lattice

At $K = 3$, mean normalized regret falls from 6.3% for global static to 2.7% for OSM (a 3.6-percentage-point or 57% relative reduction). Per-operator calibration reaches 1.7% but requires a separate search at each operator (Sec. 5.4). OSM selects the exact oracle schedule in about 58% of cases; nearly equivalent schedules can produce similarly low regret. Greedy re-evaluation has higher top-3 overlap (0.78 versus 0.75) but higher regret (3.3% versus 2.7%), showing that overlap and operator-visible regret are distinct criteria.

Table 3. Normalized operator-visible regret (%) and top-3 oracle-set Jaccard at $K = 3$. Bold marks best among methods without test-operator schedule-outcome search. Test-operator access differs (Supp. Table 9).

Method	Mean ↓	p90 ↓	Jaccard ↑
Global static	6.3%	13.2%	0.62
Gradient	6.0%	12.7%	0.65
H-aware gradient	5.0%	9.9%	0.70
Hessian/Fisher	4.8%	9.4%	0.67
InfoQ-style	5.1%	9.0%	0.71
Intervention	4.4%	8.8%	0.72
Exact Shapley	3.8%	7.4%	0.76
Greedy re-eval.	3.3%	6.9%	**0.78**
OSM	**2.7%**	**6.4%**	0.75
Per-operator calib.	1.7%	3.5%	0.84
Exact oracle	0.0%	0.0%	1.00

Table 4. In-study ProxSCI-8 reconstruction quality under simulated P1. W8 A4/A8 means 8-bit weights and stagewise 4- or 8-bit activations. Mixed policies use $K = 3$. Other published SCI systems use different architectures and evaluators (Supp. Table 16).

Method	Precision	PSNR ↑	SSIM ↑
Uniform low	W8A4	34.74	0.964
Global static	W8 A4/A8	35.06	0.966
H-aware gradient	W8 A4/A8	35.16	0.967
OSM	W8 A4/A8	35.29	0.968
Per-operator static	W8 A4/A8	35.23	0.968–0.969
Uniform high	W8A8	35.43	0.969

5.3. Reconstruction quality and generalization

At $K = 3$, lower operator-visible regret need not imply higher PSNR: per-operator calibration has lower D_H regret than OSM (1.7% versus 2.7%) but slightly lower PSNR (35.23 versus 35.29 dB; Table 4). D_H uses unclipped predictions and ignores null-space error; PSNR uses clipped reconstructions and includes it. We do not interpret the 0.06 dB difference without paired uncertainty. OSM improves on the global-static PSNR by 0.23 dB and remains 0.14 dB below uniform W8A8.

Across PTQ calibration rules and QAT, OSM regret ranges from 2.3% to 2.7% (Supp. Table 13). Activation precision exhibits stronger operator-conditioned reordering (WORI 0.25–0.34) than weight precision (0.14–0.20). Transfer is strongest for held-out mask seeds ($\tau = 0.78$–0.84; regret 2.2–2.9%); extrapolative operators give higher mean regret (3.6–5.1%). A separate non-blind deblurring study shows the same qualitative effect ($\tau = 0.70$–0.77; regret

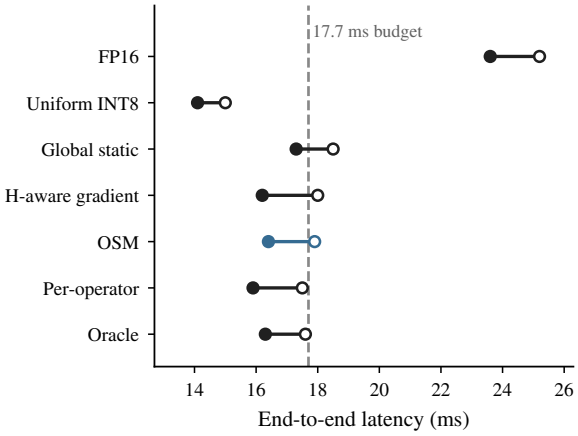


Figure 2. **Native batch-1 latency on A100 (P4).** Filled markers: p50; open markers: p95. The "Oracle" row denotes the reference schedule used in the native timing comparison. Measurements include sensing, conversion, policy selection, dispatch, and synchronization; engine construction and loading are excluded. Dashed line: latency budget (0.75× FP16 p50).

3.7–4.8%; Supp. Table 17), with a different reconstruction architecture.

Operator mismatch. Under calibrated conditions OSM regret is 2.5–3.2%; combined blur, gain, registration, and covariance mismatch raises it to 11–15% (Supp. Table 18). The allocation rule therefore requires reliable operator calibration. Global-static regret under each mismatch condition and fallback recovery were not evaluated.

5.4. Native deployment

For native deployment (P4), we separately recalibrate susceptibility for INT8↔FP16 stage execution; The P1 quality results in Table 4 do not apply to these native schedules. At the reported latency constraint of at most 0.75 times the FP16 p50 (17.7 ms), OSM measures 16.4 ms p50 (17.9 ms p95) against 23.6 ms for FP16, a ratio of 1.44×. This speed difference demonstrates native mixed-precision execution under the chosen budget, not an OSM-specific speed advantage over other allocations: H-aware gradient reaches 16.2 ms, per-operator calibration reaches 15.9 ms, and uniform INT8 reaches 14.1 ms.

For a new H, measured OSM *score/setup* time is 37.8 ms median (46.5 ms p90), whereas exhaustive per-operator search over the 256 schedules on the fixed calibration subset is 286 s median (IQR 249–341 s). Engine construction and loading are excluded from the 37.8 ms setup measurement. With cached per-snapshot latencies of 16.4/15.9 ms, the measured full-search cost gives an illustrative crossover of approximately 5.6×10^5 eight-frame SCI snapshots. This is about 5.2 h at 30 *snapshots*/s, but about 41.5 h if the input video runs at 30 frames/s (one snapshot per eight input frames). The estimate excludes engine build/loading and is sensitive to the competitor's search strategy; a 56-schedule $K = 3$ search or cheaper operator-aware baseline may change the crossover substantially.

5.5. Calibration, interactions, and policy stability

Increasing the calibration set improves held-out agreement: held-out Kendall τ rises from 0.64–0.68 at 256 calibration tuples to 0.78–0.84 at 2,048, while mean regret falls from 4.7–5.5% to 2.2–2.9%. Low-rank factors are selected from $\{8, 16, 32, 64, 96\}$ by joint energy and validation-order retention; most stages select 32 or 64. Separated calibration sub-pools yield stage-score $\tau = 0.75$–0.88 and shift held-out regret by 0.3–1.2 percentage points.

On the reported 256-schedule lattice, an additive fit accounts for mean in-sample $R^2 = 0.84$ and an additive-plus-pairwise fit for $R^2 = 0.93$; these are descriptive fits rather than out-of-sample predictive guarantees. Additional seed-level tables are provided in the supplement.

Real-measurement scope. Public real-SCI examples lack aligned ground-truth high-speed video, so no quantitative real-data PSNR or SSIM is claimed here. The main reported quantitative results come from controlled simulated SCI and the separately instrumented native timing study.

6. Discussion

Geometry-dependent allocation. Changing the measurement operator alters which quantization perturbations are visible, even for a fixed trained solver and matched reconstruction difficulty. Frozen-stage and full-solver evaluations show the induced stage-order changes, and the exact lattice establishes their policy consequences. The susceptibility score estimates this geometry-sensitive ordering while leaving residual interactions to the measured schedule evaluation.

When to use OSM. For a sensor with stable geometry, a global-static policy avoids repeated selection; when an expensive per-operator search can be amortized across many reconstructions, per-operator calibration can achieve lower operator-visible regret. OSM is suited to calibrated sensors that are reconfigured frequently enough to make repeated exhaustive searches costly. The measured selection/setup costs are 37.8 ms for OSM and 286 s for exhaustive per-operator search; the operating-point crossover depends on operator lifetime, alternative search procedures, and engine-build costs excluded from the reported setup timings. Simulated regret and native steady-state latency remain separate comparisons.

Approximation and resources. The self-energy score omits residual alignment and stage interactions; additive and pairwise fits explain $R^2 = 0.84$ and 0.93 in sample. Low-rank factor storage scales as $O(n \sum_j \ell_j)$, motivating explicit memory accounting when scaling to larger networks.

7. Limitations

D_H is teacher-relative and blind to null-space error; ground-truth PSNR and SSIM are evaluated separately. The analysis assumes calibrated linear sensing and known noise covariance; under combined reported mismatch, OSM regret rises to 11–15%, and an uncertainty-triggered fallback has not been evaluated. Susceptibility depends on the model, quantizer, and calibration distribution, and the OSM self-energy score does not include the cross-stage residual-alignment term in exact singleton gain. Exact enumeration is feasible for the eight-stage diagnostic model; generalization to substantially larger solvers remains to be established. Simulated W8A4/W8A8 reconstruction quality and native INT8/FP16 timing are distinct experiments, not a matched accuracy–latency frontier. Native gains depend on the device and engine configuration; sufficiently long-lived fixed sensors can favor per-operator calibration.

8. Conclusion

A change in sensing geometry can reorder precision-sensitive stages and alter the optimal mixed-precision schedule for a fixed inverse model. OSM uses calibrated perturbation statistics and the current operator to select precision without a new per-operator search. Exact schedule enumeration quantifies the effect for calibrated operators; separate A100 timing shows that the selected mixed-precision schedule runs 1.44× faster than FP16 under the tested budget. More broadly, the importance of a numerical approximation in an inverse solver depends on the physical operator through which its error is observed.

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